

# Three-Month Research Plan for ICRA 2027 Publications

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**Abstract**—This 3-month plan targets dual publications: **Paper 1** on VLA+ESN for real-time humanoid control (ICRA 2027), and **Paper 2** on robotics security (RA-L 2026 or NeurIPS 2025). Both tracks include detailed timelines, deliverables, and risk mitigation.

## I. EXECUTIVE SUMMARY

- **Track A (Primary):** ICRA 2027 - Paper 1 (Sep 2026 submission), requires on-site attendance (May 2027, Seoul)
- **Track B (Backup):** NeurIPS 2025 - Paper 1 (Sep 2025 submission), online presentation OK
- **Track C (Backup):** ICML 2025 - Paper 1 (Aug 2025 submission), online presentation OK
- **Primary author:** Andrew. **Co-author:** Tariq Abdul Quddoos (PVAMU)

## II. PAPER 1: CLOSING THE FREQUENCY GAP

*Detail plan:* research\_paper1\_plan.tex (revised 5 Aug 2026). Summary only below.

### A. Sanity Check (Measured vs. Missing)

- **Done:** UnifoLM gap 573.2 ms / 1.745 Hz (PyTorch) and 593.7 ms (Nsight); ESN ridge  $9.83 \times 10^{-4}$  rad; dual-process 248 Hz under *mock* VLA; MuJoCo *oracle* joint RMSE  $1.37 \times 10^{-3}$  rad (not a 17% success rate).
- **Missing (submission-critical):** live UnifoLM closed-loop vs. ZOH/linear; filled ablation/baseline tables; camera-ready figures.
- **Runway:** ~6 weeks to 15 Sep 2026—not a 12-week plan.

### B. Motivation (one line)

Keep language-conditioned VLA intents; add a frozen ESN as the dynamical interface that closes the measured  $\sim 57\text{--}59\times$  rate gap and must beat ZOH/linear on success and jerk.

### C. Critical Path (6 Weeks)

- 1) Aug 5–14: live UnifoLM in sim (no mock) + ZOH/linear/ESN baselines
- 2) Aug 15–31: comparative trials ( $\geq 20$ /method wipe) +  $N/p$  ablations; G1 best-effort
- 3) Sep 1–12: figures + 6+1 page draft + Tariq review
- 4) Sep 13–15: PaperPlaza submit

### D. Required Citations (Top)

- 1) OpenVLA (arXiv:2503.08703, 2025)
- 2) UnifoLM-VLA (Unitree, 2026)
- 3)  $\pi_0$  / flow-matching VLA preprints (2026)
- 4) Jaeger echo-state hypothesis (2001)
- 5) Unitree G1 Edu; MuJoCo

## III. PAPER 2: ROBOTICS SECURITY

### A. Concept

ESN-based anomaly detection between VLA outputs and G1 actuators. Detects adversarial perturbations before physical harm.

### B. Threat Model (G1 Edu)

- Perception: Camera spoofing, adversarial patches
- Network: FMX DDS sniffing
- AI: Prompt injection, jailbreaks
- Physical: Joint torque injection

### C. Recommend{ed} Venues

- **Primary:** IEEE RA-L 2026 (Jun 2026, no on-site) - fits robotics+security
- **Secondary:** RSS 2026 (Oct 2026, no on-site) - excellent robotics venue
- **Alternative:** ACM/IEEE CSS 2027 (Mar 2027, no on-site) - security focus

### D. 10-Week Timeline (Paper 2)

| Week | Task                                | Owner        |
|------|-------------------------------------|--------------|
| 1-2  | Lit review                          | Osemu        |
| 3    | Threat model                        | Both         |
| 4-6  | Experiments (attacks + ESN defense) | Tariq (lead) |
| 7-8  | Draft                               | Osemu        |
| 9    | Review                              | Both         |
| 10   | <b>Submit</b>                       | Both         |

## IV. COLLABORATION WITH TARIQ ABDUL QUDDOOS

- **Paper 1:** Andrew (primary), Tariq (co-author)
  - Andrew: Architecture, experiment design, writing
  - Tariq: Code implementation, experiment execution, validation
- **Paper 2:** Both 50-50 or You 60-40 (depending on his interest)
- **Compensation:** PVAMU CREDIT Center covers costs
- **Ethical:** Tariq will actively participate; no fabrication

A. Contact Tariq (this week)

Subject: Invitation to Co-Author Robotics Research

Dear Tariq,

I hope you're well. I'm Osemu, a CREDIT Center

Paper 1: "Closing the Frequency Gap" - VLA + ESN-based

Paper 2: "Robust Embodied AI Security" - ESN-based

Your expertise in (his research area) would be valuable.

Interested? Let me know next week.

Best,  
Andrew

V. RISK MITIGATION

**ICRA (No Show Risk: CRITICAL** - Must physically attend Seoul. If you can't travel, pivot to NeurIPS/ICML 2025.

- **Live Trial Failure:** Run 100+ trials with extra margin
- **Missing Citations:** Cross-check 20+ papers, verify DOIs
- **Tariq Unavailable:** Have external co-author backup (e.g.,

**Paper Rejection:** Both papers prepared for backup venues

VI. REFERENCES (SAMPLE)

REFERENCES

[1] **OpenVLA Team.** "OpenVLA: Open-Source Vision-Language-Action Models." *arXiv preprint arXiv:2503.08703*, 2025.  
[2] **H. Jaeger.** "The 'Echo State' Hypothesis." *J. Comp. Sci. Eng.*, 2001.  
[3] **Unitree Robotics.** "G1 Edu User Manual." 2024.  
[4] **Todorov, G.** "MuJoCo: A Fast and Diverse Physically-Based Simulator." *arXiv:2105.04663*, 2024.